ANALYTICAL METHODS:

1 DESCRIPTION

1.1 Physically examine a representative sample of the material for Appearance and other characteristics: it is White or almost white, crystalline powder or colourless crystals or white or almost white pearls. Freely soluble in water, practically insoluble in anhydrous ethanol**.**

2 IDENTIFICATION

2.1 SODIUM TEST

2.1.1 Potassium Antimonate V

2.1.2 Weigh about 2g of potassium antimonite V into a 250ml beaker.

2.1.3 Add 95ml of hot water, drop a bar magnet and mix (using a magnetic stirrer) till dissolution is complete

2.1.4 Cool rapidly to room temperature in a water-bath at 25°C

2.1.5 Add 50ml of 5%w/v solution of potassium hydroxide and 1ml of 2M sodium hydroxide and mix.

2.1.6 Allow standing for 24 hours, filter through Whatman #1 filter-paper into another 250ml beaker .Dilute to 150ml with distilled water and mix.

2.2 Procedure

2.2.1 Accurately weigh and transfer about 0.5g of the sample into a suitable porcelain crucible

2.2.2 Add 10ml of concentrated nitric acid and swirl to moisten every grain with the acid.

2.2.3 Transfer crucible and content onto water-bath and evaporate to dryness.

2.2.4 Transfer into a furnace operating at 800°C and ignite the residue until it becomes white

2.2.5 Add 5ml of distilled water, gently swirl till dissolution is complete. **Label Solution S.**

2.2.6 Filter the solution through conically-folded Whatman #1 filter-paper into a 50ml beaker (**keep solution for all the identification tests!)**

2.2.7 Pipette 2ml of the solution into a test tube.

2.2.8 Add 2ml of % 15w/v solution of potassium carbonate and heat to boiling; **No precipitate is produced.**

2.2.9 Add 4ml of potassium antimonate V and heat to boiling.

2.2.10 Allow cooling in ice and if necessary scratch the inside of the tube with a glass rod; **a dense, white precipitate is produced.**

2.3 CHLORIDE TEST

2.3.1 Reagents

2M nitric acid

Pipette 12.6ml of concentrated Nitric acid to 100ml volumetric flask containing 60ml distilled water, dilute to the mark with distilled water and mix

4.25% w/v silver nitrate

Accurately weigh and transfer about 4.25g silver nitrate to 100ml volumetric flask, dilute to the mark with distilled water and mix

10M ammonia

Dilute about 41g of con ammonia to 100ml volumetric flask, dilute to the mark with distilled water and mix

2.3.2 Sample preparation

2.3.3 Dissolve 1g of sample in 2 ml of distilled water.

2.3.4 Acidify with 2M nitric acid, add 0.4 ml of 4.25% w/v silver nitrate solution.

2.3.5 Shake and allow to stand; **a white precipitate is produced**

2.3.6 Filter and wash the precipitate with three 1-ml quantities of water.

2.3.7 Carry out this operation rapidly in subdued light, disregarding the fact that the supernatant solution may not become perfectly clear: Suspend the precipitate in 2 ml of water and add 1.5ml of 10M ammonia; **the precipitate dissolves easily with the possible exception of a few large particles which dissolve slowly.**

3 APPEARANCE OF SOLUTION

3.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

3.2 Add 60ml of freshly distilled and cooled water; mechanically shake/sonicate till dissolution is complete.

3.3 Dilute to the mark with distilled water and mix. **Solution S is clear and colourless**

3.4 Observe the solution against bright northerly light: Dry for 2 hours.

4 IRON

4.1 20ppm Iron Standard Stock

4.1.1 Accurately weigh and transfer about 140.4mg of Ferrous Ammonium Sulphate hexahydrate $Fe(NH_4)_2(SO_4)_6H_2O$] into a one-litre volumetric flask.

4.1.2 Add 300ml of distilled water, 25ml of concentrated sulphuric acid, mechanically shake/sonicate till dissolution is complete.

4.2 Sample Preparation

4.2.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

4.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

4.2.3 Dilute to the mark with same solvent and mix.

4.3 Procedure

4.3.1 Pipette 10.0ml each of the sample and standard preparations into separate 25ml volumetric flask.

4.3.2 Add 2ml of 20%w/v citric acid, 0.1ml of mercpatoacetic acid, mix and compare sample and standard preparations

4.3.3 Dilute to the mark with distilled water and mix.

4.3.4 Using a suitable measuring cylinder, transfer 20ml of each preparation into separate Nessler cylinders and compare the intensity of the colour developed: **Pink colouration obtained in the test solution is not more intense than the standard solution.**

5 ACIDITY OR ALKALINITY

5.1 **Reagents**

5.1.1 0.1%w/v bromothymol blue indicator ; ; Accurately weigh and transfer about 01g bromothymol blue to 100ml volumetric flask, dilute to the mark with distilled water and mix

5.1.2 0.01M hydrochloric acid; dilute 10ml of 0.01M hydrochloric acid to 100ml with distilled water

5.1.3 0.01M sodium hydroxide ; dilute 10ml of 0.1M sodium hydroxide to 100ml with distilled water

5.2 Sample Preparation

5.2.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

5.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

5.2.3 Dilute to the mark with same solvent and mix.

5.2.4 Pipette 20.0ml of this solution into a 100ml conical flask; add 0.1ml of 0.1%w/v bromothymol blue indicator.

5.2.5 Titrate with 0.01M hydrochloric acid or 0.01M sodium hydroxide solution to a change in the colour of the indicator: **Not more than 0.5 ml of 0.01M hydrochloric acid or 0.01M sodium hydroxide is required to change the colour of the indicator.**

6 BROMIDES

6.1 Phenol Red Solution

6.1.1 Accurately weigh and transfer about 0.105g of phenol red into 100ml volumetric flask

6.1.2 Add 3.0ml of 0.1M sodium hydroxide, 25ml of 96% ethanol and mechanically shake/sonicate till dissolution is complete.

6.1.3 Dilute to the mark with distilled water and mix.

6.2 Standard Preparation

6.2.1 Accurately weigh and transfer about 15mg of potassium bromide into a 1000ml volumetric flask.

6.2.2 Add 600ml of distilled water and mechanically shake/sonicate till dissolution is complete.

6.2.3 Dilute to the mark with distilled and mix.

6.3 Sample Preparation

6.3.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

6.3.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

6.3.3 Dilute to the mark with distilled water and mix.

6.4 Procedure

6.4.1 Pipette 1.0ml each of the sample and standard preparations into separate 10ml volumetric flasks.

6.4.2 Add to each 10ml flask, 6.0ml of distilled water, 2.0ml of phenol red solution, 1.0ml of 0.01%w/v chloramines solution and mix immediately.

6.4.3 After exactly 2 minutes, add to each 10ml flask, 0.15 ml of 0.1M sodium thiosulphate and mix.

6.4.4 Using a suitable spectrophotometer and 1-cm cells, measure the absorbance of the sample and standard preparations at 590nm against distilled water: **The absorbance of the sample solution is not greater than that of a standard preparation (50 ppm).**

7 IODIDES

7.1 Starch Solution, Iodide-free

7.1.1 Weigh about 1g of soluble starch into 250ml beaker

7.1.2 Add 5ml of distilled water and and triturate with glass rod.

7.1.3 With continuous stirring, 100ml of boiling water.

7.1.4 After the addition of the boiling water, continue to stir for further 5 minutes.

7.2 Sodium Nitrite-Sulphuric Mixture **(Prepare 10 mniutes before use!!!)**

7.2.1 Mix 1.0ml of 1.5%w/v sodium nitrite, 2ml of 0.5M sulphuric acid, 25ml of iodide-free starch solution and 25ml of distilled water

7.3 Procedure

7.3.1 Accurately weigh about 5g of sample into a 250ml beaker

7.3.2 Moisten the sample by the dropwisely adding the freshly prepared sodium nitrite-sulphuric mixture

7.3.3 Allow 5 minutes and examine the sample in daylight: **The substance shows no blue colour.**

8 MAGNESIUM AND ALKALI METALS

8.1 **Reagents**

8.1.1 0.01M disodium edetate prepared.

8.1.2 0.1M zinc sulphate prepared.

8.2 SAMPLE PREPARATION

8.2.1 Into a 500ml beaker, measure (with a cylinder), 200ml of distilled water.

8.2.2 Add 0.1 g of hydroxylamine hydrochloride, 10 ml of ammonia buffer pH 10.0, 1 ml of 0.1M zinc sulphate and about 15 mg of mordant black 11 triturate..

8.2.3 Heat to about 40° and titrate with 0.01M disodium edetate VS until the violet colour changes to a full blue.

8.2.4 Accurately weigh and transfer about 10g of sample into the beaker and mix until dissolution is complete.

8.2.5 If the colour of the solution changes to violet, titrate with 0.01M disodium edetate VS until the full blue colour is again produced.

8.2.6 The volume of 0.01M disodium edetate VS used is not more than 2.5ml

9 Aluminium

9.1 Acetate Buffer pH 6.0

9.1.1 Weigh about 40g of ammonium acetate into a 250ml beaker

9.1.2 Add 100ml of distilled water, 1.6ml of glacial acetic acid and mix till dissolution is complete.

9.1.3 Adjust the pH to 6.0, if necessary, using 10M ammonia or 5M acetic acid

9.1.4 Dilute to the 200ml- mark with distilled water and mix.

9.2 200ppm Aluminium Standard Solution

9.2.1 Weigh about 0.352g of [Aluminium Potassium Sulphate ($AlK(SO_4)_2,12H_2O$) into a 100ml volumetric flask.

9.2.2 Add 30ml of distilled water, 10 ml of 1M sulphuric acid and mechanically shake/sonicate till dissolution is complete.

9.2.3 Dilute to the 100ml- mark with distilled water and mix. **Store this solution for not more than 3 months.**

9.3 2ppm Aluminium Standard Solution (Prepare Immediately Before Use!)

9.3.1 Pipette 1.0ml of 200ppm aluminium solution into 100ml volumetric flask.

9.3.2 Dilute to the 100ml- mark with distilled water and mix. Store this so

9.4 0.5% w/v Solution Of 8-Hydroxyquinoline

9.4.1 Weigh and transfer about 1g of 8-hydroxyquinoline into a dry 200ml volumetric flask.

9.4.2 Add 100ml of chloroform, mechanically shake/sonicate till dissolution is complete.

9.4.3 Dilute to the 200ml- mark with chloroform and mix.

9.5 SAMPLE PREPARATION

9.5.1 Accurately weigh and transfer about 4g of sample into a 100ml volumetric flask.

9.5.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

9.5.3 Add 10ml of acetate buffer pH 6.0, dilute to the mark with distilled water and mix.

9.5.4 Pipette 20ml of this solution into 100ml beaker.

9.6 Procedure

9.6.1 Into three 250 ml Separate funnels, pipette and add in the following order:

Reagent s	Sample	Standard	Blank
Sample Solution	100.0ml	-	-
2ppm Aluminium Standard Solution	-	2.0ml	-
Water	-	98.0ml	100.0ml
Acetate Buffer pH 6.0	10.0ml	10.0ml	10.0ml

9.6.2 Separate extract each of the sample, standard and blank preparations in the above table with successive quantities of 20, 20 and 10 ml of a 0.5% w/v solution of 8-hydroxyquinoline in chloroform, filtering the extracts through chloroform-wet

Whatman #1 filter paper into separate, 'sample', 'standard', 'blank' labeled, dry 50ml volumetric flasks

9.6.3 Using a suitable spectrophotometer and 1-cm cells, measure the absorbance of the test solution and that of the standard solution against the prepared blank at wavelength of 392 nm and 518 nm: **The absorbance of the test solution is not greater than that of the standard solution.**

10 BARIUM

10.1 Pipette 5.0ml of solution S and 5ml of distilled water into test tube and mix:

10.2 Add 2ml of 1M sulphuric acid and mix

10.3 After at least 2 hours, any opalescence in the sample solution is not more intense than that in a mixture of 5ml of solution S and 6ml of distilled water

11 FERROCYANIDES

11.1 Ferric Ammonium Sulphate

11.1.1 Using a measure cylinder, measure 50ml of distilled water into a 100ml volumetric flask.

11.1.2 Carefully add, by pipette and through the wall, 1.5ml of concentrated sulphuric acid.

11.1.3 Cool flask and content to room temperature, with mixing, under running tap water

11.1.4 Accurately weigh and transfer about 50mg of Ferric Ammonium Sulphate, into the above flask, mechanically shake/sonicate till dissolution is complete.

11.1.5 Dilute to the mark with distilled water and mix.

11.2 <u>Ferrous Sulphate</u>

11.2.1 Accurately weigh and transfer about 0.950g of Ferric Ammonium Sulphate, into the above flask, mechanically shake/sonicate till dissolution is complete.

11.2.2 Dilute to the mark with distilled water and mix.

11.3 <u>Procedure</u>

11.3.1 Accurately weigh and transfer about 2g of sample into a 100ml beaker.

11.3.2 Add 6ml of distilled water and mix till dissolution is complete.

11.3.3 Transfer the ferric ammonium sulphate solution and the ferrous sulphate solution into a 250ml beaker and mix for 5 minutes

11.3.4 Pipette 0.5ml of mixture in step 'iii' to the sample solution in step 'ii' and mix: **No blue colour develops within 10 minutes.**

12 SULPHATES.

12.1 1000ppm SULPHATE STANDARD PREPARATION

12.1.1 Accurately weigh and transfer about 0.181g of analytic grade potassium sulphate into a 100ml volumetric flask.

12.1.2 Add 60ml of 30% v/v ethanol in water and sonicate till dissolution is complete.

12.1.3 Dilute to the mark with 30% v/v ethanol in water and mix "(1000ppm sulphate standard stock solution).

12.1.4 Pipette 15.0ml of this solution into another 100ml volumetric flask, dilute to the mark with 30% v/v ethanol in water and mix (150ppm sulphate working standard).

12.2 SAMPLE PREPARATION

12.2.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

12.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

12.2.3 Dilute to the mark with same solvent and mix.

12.3 Procedure

12.3.1 Pipette 15.0ml each of this solution and 1ml of acetic acid into 25ml volumetric flasks.

12.3.2 To another flask, 13.0ml of distilled water, 2ml of 150ppm sulphate standard solution, and 1ml of acetic acid into 25ml volumetric flasks.

12.3.3 To each flask, add 1ml of 25% w/v solution of barium chloride solution and 1ml of 2M nitric acid Dilute to the mark with 30% v/v ethanol in water and mix.

12.3.4 . Shake each flask and allow standing for 5 minutes: **Any opalescence produced is not more intense than that of Sulphate standard (150ppm SO_4).**

13 **LOSS ON DRYING**

13.1 Weigh two labelled petri dishes (W1) that has previously been dried at 105°C and cooled

13.2 Accurately weigh about 1g onto a petri-dish, ensuring they are evenly spread, then determine their weight together (W2).

13.3 Transfer into an Oven operating at 105°C.

13.4 Dry for 2 hours.

13.5 Transfer into a desiccators to cool room temperature.

13.6 Weigh the petri-dish with its content (W3).

13.7 Calculate the moisture content as follows:

$$\% \text{ Moisture} = \frac{(W2 - W3) \times 100}{W2 - W1}$$

Where,

W1 = weight of empty petri-dish, g)

W2 = weight of sample and petri-dish, g.

W3 = weight of petri-dish and the dried sample, g

14 NITRITES

14.1 Sample Preparation

14.2 Accurately weigh and transfer about 2g of sample into a 20ml volumetric flask.

14.3 Add 10ml of distilled water and mechanically shake/sonicate till dissolution is complete.

14.4 Dilute to the mark with distilled water and mix.

14.5 Using a suitable spectrophotometer and 1-cm cells, measure the absorbance at 354nm against water as blank: , The solution complies with the limit test for nitrites (25 ppm).

15 PHOSPHATES

15.1 Tin(II) Chloride Solution (Stannous chloride solution)

15.1.1 Accurately weigh and transfer about 20g of Tin into a 250ml beaker.

15.1.2 Add 85ml of concentrated hydrochloric acid.

15.1.3 Transfer onto a hotplate 9inside fume-hood) and heat until dissolution is complete and until no more hydrogen is being evolved.

15.1.4 Cool to room temperature and mix (Filter if necessary). **Store over an excess of tin and protected from air.**

15.2 Working Tin(II) Chloride Solution

15.2.1 Immediately before use, pipette 9.0ml of Tin (II) Chloride Stock Solution, dilute to the mark with 2M hydrochloric acid and mix.

15.3 Sulphomolybdic Reagent

15.3.1 Using a 50ml measuring cylinder, add 20ml of distilled water into a 250ml beaker.

15.3.2 Weigh and add 2.5g of ammonium molydate into the beaker

15.3.3 Transfer onto a hotplate and heat until dissolution is complete.

15.3.4 Allow cooling for about 5 minutes, then cool to room temperature in cold water bath.

15.3.5 Into another 250ml beaker, separately, add with care, 28 ml of sulphuric acid to 50 ml of water and cool to room temperature

15.3.6 Mix the two solutions together and store in plastic container.

15.4 Standard Phosphate Stock Solution

15.4.1 Accurately weigh and transfer about 71.6mg of potassium dihydrogen orthophosphate into a 100ml volumetric flask.

15.4.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

15.4.3 Dilute to the mark with distilled water and mix.

15.5 Sample Preparation

15.5.1 Accurately weigh and transfer about 0.4g of sample into a 100ml volumetric flask.

15.5.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

15.5.3 Dilute to the mark with distilled water and mix.

15.6 Procedure

15.6.1 To each of the sample and the 5ppm standard preparations, add 4 ml of sulphomolybdic reagent and shake for 1 minute.

15.6.2 Add 0.1 ml of tin(II) chloride working solution, allow standing for 10 minutes and examine 20 ml of the resulting solution. **Any colour produced is not more intense than that produced in the reference solution.**

16 **ARSENIC**

NOTE; Arsenic is very poisonous therefore adequate measure should be taken when using this equipment

A . ensure adequate ventilation

B Avoid breathing in the vapour (use fume cupboard)

C Do not attempt clean-up without minimum PPE.(personal protective equipment)

16.1 9.5%w/v Lead Acetate Solution

16.1.1 Accurate weigh and transfer about 9.5g of Lead Acetate into a 100ml volumetric flask

16.1.2 Add 60ml of distilled water, mechanically shake and sonicate till dissolution is complete.

16.1.3 Dilute to the mark with distilled water and mix. Store in an air-tight container

16.2 Preparation Of Lead Acetate Cotton

16.2.1 Transfer 10ml of lead acetate solution and 1ml of 2M acetic acid into a 250ml beaker

16.2.2 Cut off several pieces of absorbent cotton, each weighing about 60mg.

16.2.3 Immerse several small pieces of absorbent cotton in a lead acetate-acetic acid mixture.

16.2.4 Drain off the excess of liquid by placing it on several layers of filter paper without squeezing the cotton

16.2.5 Allow drying at room temperature and Store in an airtight container.

16.3 5%w/v Merucry (II) Bromide Solution

16.3.1 Accurate weigh and transfer about 5g of Merucry (II) Bromide into a 100ml volumetric flask

16.3.2 Add 60ml of absolute ethanol, mechanically shake and sonicate till dissolution is

complete.

16.3.3 Dilute to the mark with ethanol and mix. Store in an air-tight container

16.4 Mercury(II) Bromide Paper

16.4.1 From one circle of Whatman #1 filter paper, cut a piece measuring 200mm by 15mm

16.4.2 Fold it in two and immerse it in 10ml of mercury (II) bromide solution in a 50ml beaker

16.4.3 Decant the excess liquid and allow the papers to dry, protected from light, by suspending them over a non-metallic thread

16.4.4 Cut away the folded edges to a width of 1cm and similarly remove the outer edges

16.4.5 Cut the remaining strips into 15-mm squares or discs of 15mm diameter.

16.4.6 Store in a glass-stoppered container wrapped with black paper.

16.5 Tin(II) Chloride Solution

16.5.1 Accurately weigh and transfer about 20g of Tin (II) Chloride (Stannous Chloride) into a 250ml beaker

16.5.2 Add 85ml of concentrated hydrochloric acid. **Please handle in the fume-hood!!!**

16.5.3 Transfer onto a hotplate and heat until evolution of hydrogen ceases.

16.5.4 Allow cooling to room temperature and transfer into a amber coloured reagent bottle and tightly cover, protected from air..

16.6 STANDARD PREPARATION

16.6.1 Accurately weigh and transfer about 264mg of arsenic trioxide into 200ml volumetric flask.

16.6.2 Add 5ml of 2M sodium hydroxide and mechanically shake/sonicate till dissolution is complete.

16.6.3 Dilute to the mark with distilled water and mix (10ppm). **Store I amber bottle for $\leq$ 3 months.**

16.6.4 Pipette 10.0ml of this solution into 100ml volumetric flask, dilute to the mark with distilled water and mix (1ppm). **Prepare immediately before use!**

16.6.5 Pipette 5.0ml of the 1ppm arsenic standard solution into each 250ml iodine-flask

16.6.6 Add 5ml of 95-98% sulphuric acid, 10ml of 30% hydrogen peroxide, heating until copious fumes of sulphur trioxide forms.

16.6.7 Carefully add 10ml of water, mix and heat further till traces/fumes of hydrogen peroxide are removed.

16.6.8 Allow cooling, rinse the walls of the flask and dilute with water to make 20ml of solution.

16.7 SAMPLE PREPARATION

16.7.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

16.7.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

16.7.3 Dilute to the mark with distilled water and mix.

16.8 Procedure

16.8.1 *To the sample and standard preparations in 100ml quick-fit conical flask, add 15ml of concentrated hydrochloric acid, 2ml of 16.5%w/v potassium iodide, 0.1ml Tin (II) chloride solution in concentrated hydrochloric acid and mix.*

16.8.2 While waiting insert between 50 – 60mg of lead acetate cotton loosely packed into the lower tube of the Arsenic Apparatus.

16.8.3 Place a small square of circular mercury II Bromide paper between the surface of the tube large enough to cover the orifice of the tube.

16.8.4 Immediately assemble the two parts of the apparatus.

16.8.5 Immerse the flask in water at about 90°C and leave for 2hrs.

16.8.6 Mercury II Bromide paper is stained.

16.8.7 Repeat the above using in place of the sample 1ml of Arsenic standard solution (1ppm As) + 24ml of distilled water: **Any stain produced by sample is not more intense than that of the standard (1ppm As)**

17 **POTASSIUM**

17.1 500ppm Standard Potassium Stock Solution

17.1.1 Accurately weigh and transfer about 222mg - 224mg f potassium sulphate into a 100ml volumetric flask.

17.1.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

17.1.3 Dilute to the mark with distilled water and mix.

17.2 Sample Preparation

17.2.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

17.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

17.2.3 Dilute to the mark with distilled water and mix.

17.3 Procedure

17.3.1 Pipette 10ml each of sample and standard preparations into separate 25ml volumetric flask.

17.3.2 Add 2ml of sodium tetraphenylborate R

17.3.3 After 5 minutes, any opalescence in the sample solution is not more intense than that of the standard solution (500ppm)

18 **ASSAY METHOD A**

18.1 Ferric Ammonium Sulphate

18.1.1 Accurate weigh and transfer about 0.2g of ferric ammonium sulphate into a 100ml volumetric flask
Add 560ml of distilled water, 5ml of concentrated nitric acid; mechanically shake/sonicate till dissolution is complete.
Dilute to the mark with distilled water and mix.

18.1.2 0.1M Silver Nitrate ; prepared as per SOP NO QCC016-05

18.1.3 0.1M Ammonium Thiocynate; prepared as per SOP NO QCC016-05

18.1.4 <u>2M nitric acid</u>
Pipette 12.6ml of conc Nitric acid to 100ml volumetric flask, dilute to the mark with distilled water and mix

18.1.5 <u>Procedure</u>
18.1.6 Accurately weigh about 100mg each into two separate 250ml conical flasks.
18.1.7 Add 50 ml of distilled water and sonicate/swirl till dissolution is complete.
18.1.8 Add 5ml of 2M nitric acid, 25.0ml of 0.1M Silver Nitrate and 2ml of dibutyl phthalate and mix for 2 minutes, using the magnetic stirrer.
18.1.9 Add 2ml of 0.2%w/v ferric ammonium sulphate as indicator and titrate with ,0.1M Ammonium Thiocyanate with vigorous mixing/shaking until there is a colour change.

Carry out a blank titration, omitting the sample.

%Assay

$$= \frac{(\text{Sample Titre - Blank Titre}) \times 5.844\text{mg} \times \text{Actual Molarity of Ammonium Thiocynate} \times 100 \times 100}{\text{Sample weight (g)} \times 1000 \times 0.1 \times (100)}$$

$$= \frac{(\text{Sample Titre - Blank Titre}) \times 5.844\text{mg} \times \text{Actual Molarity of Ammonium Thiocynate} \times 100}{\text{Sample weight (g)} \times (100)}$$

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

18.2 **ASSAY METHOD B**

18.2.1 **Reagents**

18.2.2 5%w/v potassium chromate solution Accurately weigh and transfer about 5.0g potassium chromate to 100ml volumetric flask, dilute to the mark with distilled water and mix

18.2.3 0.1M Silver Nitrate

18.2.4 **Sample preparation**

18.2.5 Accurately weigh about 50mg each into two separate 250ml conical flasks.

18.2.6 Add 50 ml of distilled water and sonicate/swirl till dissolution is complete.

19 **Microbiological Tests**

19.1 **Limulus Amebocyte Lysate LAL Test**

19.1.1 Open the seals of the test tubes of lyophilized LAL Reagents

19.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

19.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

19.1.4 Incubate the reaction mixture immediately at 37Oc for 60minutes. Avoiding vibration during incubation.

19.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

19.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

19.1.7 Read reaction tubes one by one.

19.1.8 Invert the tube in one smooth motion.

19.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by the absence of solid clot after inversion.

Testing Methodology of Bacterial Endotoxin (LAL)

19.2.1 Each specimen shall be tested in duplicate

19.2.2 NMC: Take 100µl(0.1ml) LRW

19.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

19.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

19.2.5 Add the LAL reagents as mentioned below in the table 1

19.2.6 Mix gently and incubate the tubes in heating block at 370C + 1 for 60min.

19.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 1800C and note the observations.

19.2.8 If a firm gel has formed, record the results as positive.

19.2.9 A result is negative, if an intact gel is not formed.

19.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

	Pyrogen
19.3	19.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed. 19.3.2 Withhold all food from the rabbits used during the period of the test 19.3.3 Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring. 19.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit. 19.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm. 19.3.6 If the initial temperature is less than 38.80c or more than 39.8ocexclude the animal from the test. 19.3.7 Ensure all the needles and syringe to be used are new 19.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 10c. 19.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection. 19.3.10 Any rabbits showing more than 0.20c temperature difference between two successive readings in the last three readings can be excluded from the test. 19.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes. 19.3.12 Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution. 19.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the

volume of the preparation to be injected.

19.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

19.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution

19.4 **MLT**

19.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

19.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

19.4.3 Testing to be performed under LAF of specified area

19.4.4 Arrange all the required material on the LAF before starting the test.

19.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

19.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45oC) in two petri plates for TAMC

19.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45oC) in two petri- plates for TYMC

19.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate

19.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45Oc and allow for solidification.

19.4.10 Incubate the plates at 30oC to 35OC for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 200C to 25oC for 5 days in an inverted position.